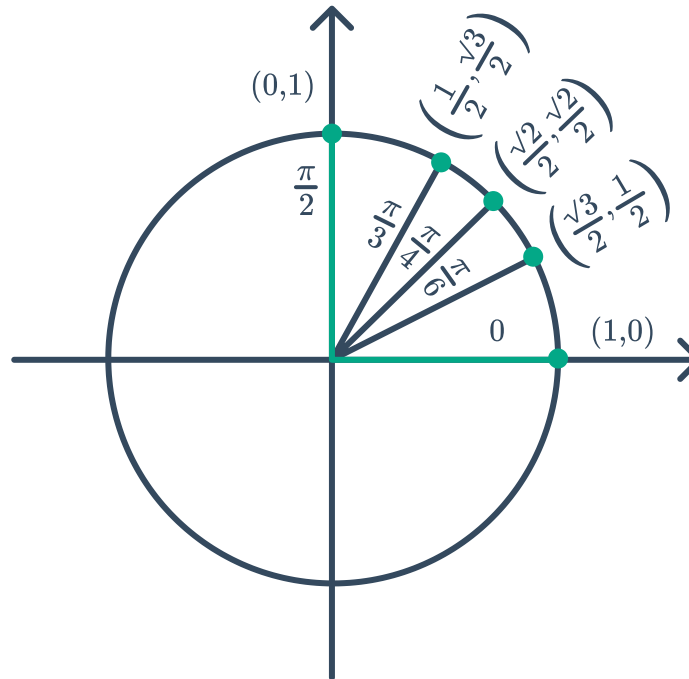


Trigonometric functions on the unit circle



1 Consider the following diagram of the unit circle:



Find the exact value of the following:

a $\cos \frac{\pi}{3}$

b $\sin \frac{\pi}{3}$

c $\sin \frac{\pi}{4}$

d $\cos \frac{\pi}{6}$

e $\cos \frac{\pi}{2}$

f $\sin \frac{\pi}{2}$

g $\cos 2\pi$

h $\tan \frac{\pi}{6}$

2 Find the exact values of the following:

a $\sin \left(\frac{7\pi}{6} \right)$

b $\cos \left(\frac{2\pi}{3} \right)$

c $\tan \left(\frac{7\pi}{4} \right)$

d $\sin \left(\frac{19\pi}{6} \right)$

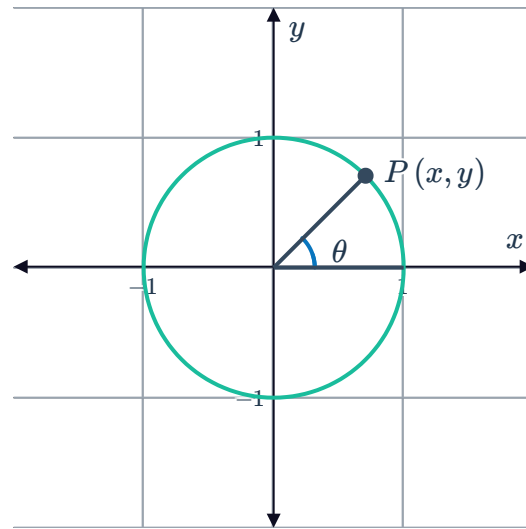
e $\cos \left(-\frac{\pi}{6} \right)$

f $\tan \left(\frac{11\pi}{4} \right)$

g $\tan \left(-\frac{\pi}{6} \right)$

h $\sin \left(-\frac{5\pi}{6} \right)$

- 3 The point P on the unit circle has the coordinates (x, y) .
Write the following in terms of x and/or y :



- a $\csc \theta$ b $\sec \theta$
c $\cot \theta$

- 4 Find the exact value of the following:

- a $\sec(-30)^\circ$ b $\cot 225^\circ$ c $\csc 240^\circ$ d $\sec 300^\circ$
e $\cot 585^\circ$ f $\csc 945^\circ$ g $\sec 945^\circ$ h $\cot 945^\circ$

- 5 Find the exact value of the following:

- a $\sec \frac{\pi}{6}$ b $\csc \frac{\pi}{3}$ c $\sec \left(\frac{2\pi}{3} \right)$ d $\csc \left(\frac{4\pi}{3} \right)$
e $\cot \left(\frac{7\pi}{6} \right)$ f $\sec \left(\frac{15\pi}{4} \right)$ g $\csc \left(-\frac{2\pi}{3} \right)$ h $\cot \left(\frac{25\pi}{6} \right)$

- 6 For each of the following functions, state:

- i The quadrants where the function is positive.
ii The quadrants where the function is negative.

- a Sine function b Cosine function
c Tangent function

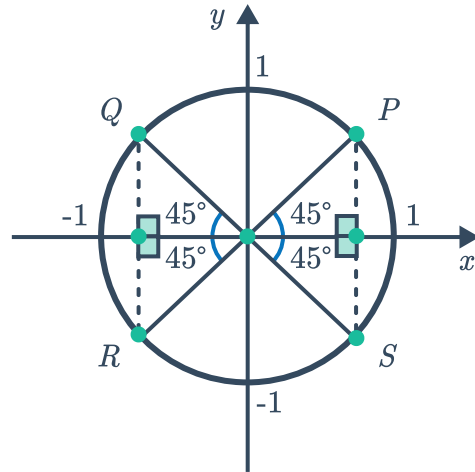
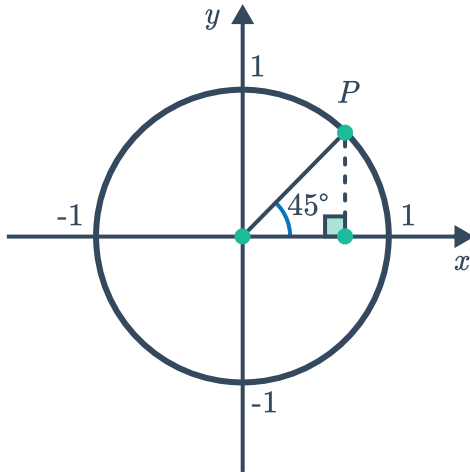
- 7 Each of the given points corresponds to the angle θ , measured counterclockwise from the positive x -axis. For each point, find the following ratios:

- i $\sin \theta$ ii $\cos \theta$ iii $\tan \theta$ iv $\csc \theta$
v $\sec \theta$ vi $\cot \theta$

- a $P \left(\frac{1}{5}, \frac{\sqrt{3}}{5} \right)$ b $P \left(-\frac{2}{5}, \frac{\sqrt{21}}{5} \right)$ c $P \left(\frac{1}{\sqrt{5}}, \frac{-1}{\sqrt{5}} \right)$



- 8 The first diagram shows a unit circle with point $P\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ marked on the circle. Point P represents a rotation of 45° counterclockwise around the origin from the positive x -axis:



- a Find the exact values of the following:
- i $\sin 45^\circ$ ii $\cos 45^\circ$ iii $\tan 45^\circ$
- b On the second diagram, the coordinate axes shows a 45° angle that has also been marked in the second, third, and fourth quadrants. For each quadrant, find the measure of the relative angle:
- i Quadrant 2 ii Quadrant 3 iii Quadrant 4
- c The points Q , R and S mark rotations of point P to the corresponding angles on the unit circle. State the exact coordinates of each point:
- i Q ii R iii S
- d Write the following in terms of an equivalent ratio of 45° :
- i $\sin 135^\circ$ ii $\cos 225^\circ$ iii $\tan 315^\circ$ iv $\sin(-45)^\circ$
- e Now, find the exact value of the following:
- i $\sin 135^\circ$ ii $\cos 225^\circ$ iii $\tan 315^\circ$ iv $\sin(-45)^\circ$

- 9 In the given diagram, $OA = 1$, and the coordinates of A are $(\cos \theta, \sin \theta)$. $\overline{CD}$ is tangent to the unit circle at A . $\triangle AOB$ is similar to $\triangle COA$ and $\triangle ODA$.

- a Show that $AC = \tan \theta$.
- b Given that $\sec \theta = \frac{1}{\cos \theta}$, show that $OC = \sec \theta$.
- c Given that $\csc \theta = \frac{1}{\sin \theta}$, show that $OD = \csc \theta$.
- d Given that $\cot \theta = \frac{1}{\tan \theta}$, show that $AD = \cot \theta$.

